

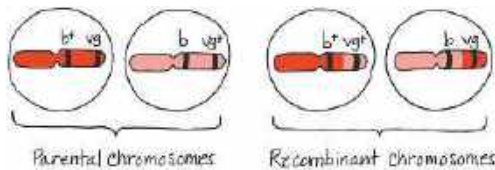
All individuals with alkaptonuria (Ariana, Benito, Elena, and Carlota) are homozygous recessive *aa*. Jorge is *Aa*, since some of his children with Ariana have alkaptonuria. Diego, Carmen, Hector, and Julio are each *Aa*, since they are all unaffected children with one affected parent. Miguel also is *Aa*, since he has an affected child (Carlota) with his heterozygous wife Carmen. Mariposa, Paloma, and Roberto can each have either the *AA* or *Aa* genotype. **14.** $\frac{1}{6}$

Chapter 15

Figure Questions

Figure 15.3 About $\frac{3}{4}$ of the F_2 offspring would have red eyes and about $\frac{1}{4}$ would have white eyes. Since the sex chromosomes are not involved in determining eye color in this hypothetical cross, about half of the white-eyed flies would be female and half would be male; similarly, about half of the red-eyed flies would be female and half would be male. (Note that the autosomes with the eye color alleles would be the same shape in the Punnett square—unlike the X and Y chromosomes—and each offspring would inherit two alleles. The sex of the flies would be determined separately by inheritance of the sex chromosomes. Thus your F_2 Punnett square would have four possible combinations in sperm and two in eggs; it would have eight squares altogether.) **Figure 15.4** The ratio would be 1 yellow round : 1 green round : 1 yellow wrinkled : 1 green wrinkled. (This is similar to a test cross.)

Figure 15.7 All the males would be color-blind, and all the females would be carriers. (Another way to say this is that $\frac{1}{2}$ the offspring would be color-blind males, and $\frac{1}{2}$ the offspring would be carrier females.) **Figure 15.9** The two largest classes would still be the offspring with the phenotypes of the true-breeding P generation flies, but now they would be gray vestigial and black normal, which is now the “parental type” because those were the specific allele combinations in the P generation (the linked alleles on their chromosomes). **Figure 15.10** The two chromosomes on the left side of the sketch below are like the two chromosomes inherited by the F_1 female, one from each P generation fly. They are passed by the F_1 female intact to the offspring and thus could be called “parental” chromosomes. The other two chromosomes result from crossing over during meiosis in the F_1 female. Because they have combinations of alleles not seen in either of the F_1 female’s chromosomes, they can be called “recombinant” chromosomes. (Note that in this example, the allele combinations on the recombinant chromosomes, b^+vg^+ and bvg , are the allele combinations that were on the parental chromosomes in the cross shown in Figures 15.9 and 15.10. The basis for calling them parental chromosomes is that they have the combination of alleles that was present on the P generation chromosomes.)



Concept Check 15.1

1. To show the mutant phenotype, a male needs to possess only one mutant allele. If this gene had been on a pair of autosomes, an individual would show the recessive phenotype only if both alleles were mutant, a much less probable situation. **2.** The law of segregation describes the inheritance of alleles for a single character. The law of independent assortment of alleles describes the inheritance of alleles for two characters. **3.** The physical basis for the law of segregation is the separation of homologs in anaphase I. The physical basis for the law of independent assortment is the alternative arrangements of all the different homologous chromosome pairs in metaphase I.

Concept Check 15.2

1. Because the gene for this eye color character is located on the X chromosome, all female offspring will be red-eyed and heterozygous ($X^{w+}X^w$); all male offspring will inherit a Y chromosome from the father and be white-eyed (X^wY). (Another way to say this is that $\frac{1}{2}$ the offspring will be red-eyed heterozygous [carrier] females, and $\frac{1}{2}$ will be white-eyed males.) **2.** $\frac{1}{4}$ ($\frac{1}{2}$ chance that the child will inherit a Y chromosome from the father and be male $\times$ $\frac{1}{2}$ chance that he will inherit the X carrying the disease allele from his mother). If the child is a boy, there is a $\frac{1}{2}$ chance he will have the disease; a female would have zero chance (but $\frac{1}{2}$ chance of being a carrier). **3.** In a disorder caused by a dominant allele, there is no such thing as a “carrier,” since those with the allele have the disorder. Because the allele is dominant, the females lose any “advantage” in having two X chromosomes, since one disorder-associated allele is sufficient to result in the disorder. All fathers who have the dominant allele will pass it along to all their daughters, who will also have the disorder. A mother who has the allele (and thus the disorder) will pass it to half of her sons and half of her daughters.

Concept Check 15.3

1. Crossing over during meiosis I in the heterozygous parent produces some gametes with recombinant genotypes for the two genes. Offspring with a recombinant phenotype arise from fertilization of the recombinant gametes by homozygous recessive gametes from the double-mutant parent. **2.** In each case, the alleles contributed by the female parent (in the egg) determine the phenotype of the offspring because the male in this cross contributes only recessive alleles. Thus, identifying the phenotype of the offspring tells you what alleles were in the mother’s (the dihybrid female’s) egg. **3.** No. The order could be A-C-B or C-A-B. To determine which possibility is correct, you need to know the recombination frequency between B and C.

Concept Check 15.4

1. In meiosis, a combined 14-21 chromosome will behave as one chromosome. If a gamete receives the combined 14-21 chromosome and a normal copy of chromosome 21, trisomy 21 will result when this gamete combines with a normal gamete (with its own chromosome 21) during fertilization. **2.** No. The child can be either $I^A I^A i$ or $I^A i i$. A sperm of genotype $I^A I^A$ could result from nondisjunction in the father during meiosis II, while an egg with the genotype ii could result from nondisjunction in the mother during either meiosis I or meiosis II. **3.** Activation of this gene could lead to the production of too much of this kinase. If the kinase is involved in a signaling pathway that triggers cell division, too much of it could trigger unrestricted cell division, which in turn could contribute to the development of a cancer (in this case, a cancer of one type of white blood cell).

Concept Check 15.5

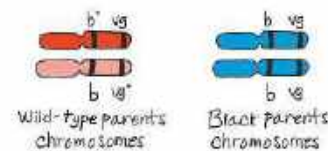
1. Inactivation of an X chromosome in females and genomic imprinting. Because of X inactivation, the effective dose of genes on the X chromosome is the same in males and females. As a result of genomic imprinting, only one allele of certain genes is phenotypically expressed. **2.** The genes for leaf coloration are located in plastids within the cytoplasm. Normally, only the maternal parent transmits plastid genes to offspring. Since variegated offspring are produced only when the female parent is of the B variety, we can conclude that variety B contains both the wild-type and mutant alleles of pigment genes, producing variegated leaves. (Variety A must contain only the wild-type allele of pigment genes.) **3.** Each cell contains numerous mitochondria, and in affected individuals, most cells contain a variable mixture of normal and mutant mitochondria. The normal mitochondria carry out enough cellular respiration for survival. (The situation is similar for chloroplasts.)

Summary of Key Concepts Questions

15.1 Because the sex chromosomes are different from each other and because they determine the sex of the offspring, Morgan could use the sex of the offspring as a phenotypic character to follow the parental chromosomes. (He could also have followed them under a microscope, as the X and Y chromosomes look different.) At the same time, he could record eye color to follow the eye color alleles. **15.2** Males have only one X chromosome, along with a Y chromosome, while females have two X chromosomes. The Y chromosome has very few genes on it, while the X has about 1,000. When a recessive X-linked allele that causes a disorder is inherited by a male on the X from his mother, there isn’t a second allele present on the Y (males are hemizygous), so the male has the disorder. Because females have two X chromosomes, they must inherit two recessive alleles in order to have the disorder, a rarer occurrence. **15.3** Crossing over results in new combinations of alleles. Crossing over is a random occurrence, and the more distance there is between two genes, the more chances there are for crossing over to occur, leading to new allele combinations. **15.4** In inversions and reciprocal translocations, the same genetic material is present in the same relative amount but just organized differently. In aneuploidy, duplications, deletions, and nonreciprocal translocations, the balance of genetic material is upset, as large segments are either missing or present in more than one copy. Apparently, this type of imbalance is very damaging to the organism. (Although it isn’t lethal in the developing embryo, the reciprocal translocation that produces the Philadelphia chromosome can lead to a serious condition, cancer, by altering the expression of important genes.) **15.5** In these cases, the sex of the parent contributing an allele affects the inheritance pattern. For imprinted genes, either the paternal or the maternal allele is expressed, depending on the imprint. For mitochondrial and chloroplast genes, only the maternal contribution will affect offspring phenotype because the offspring inherit these organelles from the mother, via the egg cytoplasm.

Test Your Understanding

1. 0; $\frac{1}{2}$; $\frac{1}{16}$ **2.** Recessive; if the disorder were dominant, it would affect at least one parent of a child born with the disorder. The disorder’s inheritance is sex-linked because it is seen only in boys. For a girl to have the disorder, she would have to inherit recessive alleles from both parents. This would be very rare, since males with the recessive allele on their X chromosome die in their early teens. **3.** 17%; yes, it is consistent. In Figure 15.9, the recombination frequency was also 17%. (You’d expect this to be the case since these are the very same two genes, and their distance from each other wouldn’t change from one experiment to another.)



4. Between T and A, 12%; between A and S, 5% **5.** Between T and S, 16%; sequence of genes is T-A-S **6.** 6%; wild-type heterozygous for normal wings and red eyes $\times$ recessive homozygous for vestigial wings and purple eyes **7.** Fifty percent of the offspring will show phenotypes resulting from crossovers. These results would be the same as those from a cross where A and B were not on the same chromosome, and you would interpret the results to mean that the genes are unlinked. (Further crosses involving other genes between A and B on the same chromosome would reveal the genetic linkage and map distances.) **8.** 450 each of blue/oval and white/round (parentals) and 50 each of blue/round and white/oval (recombinants) **9.** About one-third of the distance along the